

```
@Transactional(  
    propagation = Propagation.REQUIRED,  
    isolation = Isolation.READ_COMMITTED,  
    timeout = 5,  
    readOnly = false,  
    rollbackFor = IOException.class  
)
```

```
@Transactional  
public void createOrder() {  
  
}
```

Exceptions :

1. Runtime Exceptions
2. Checked Exceptions

Propagation :

Required

If txn exist --> use it
else --> create new

placeOrder()

audit()

Physical txn

REQUIRES_NEW

OrderService

audit

Txn1

Txn2

SUPPORTS

If txn exist --> join it
if not --> implement without Txn

MANDATORY

If txn exist --> join it
If not --> throw an exception

NOT_SUPPORTED

Always execute without a txn

txn exist --> suspend

NEVER

I want to execute without txn

txn exist --> throw exception

NESTED

txn exist -> join
not exist -> create new

save points

BEGIN

op1

op2

save point

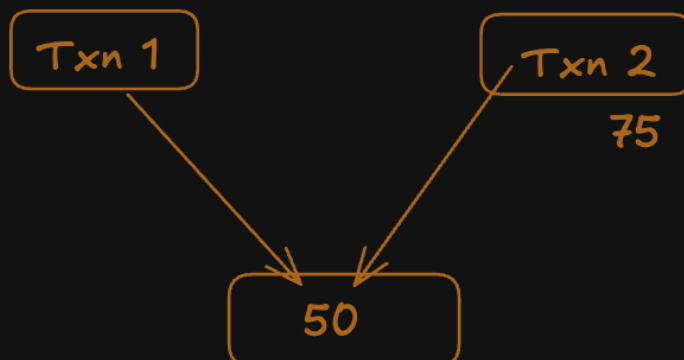
op3

op4

ROLL_BACK

Isolation Levels :

Dirty Read



Time	Transaction T1	Transaction T2
1	Begins	
2	Updates balance to ₹2,000	
3		Begins
4		Reads balance as ₹2,000
5	Rolls back	
6	Balance returns to ₹10,000	Continues using ₹2,000

10,000 balance

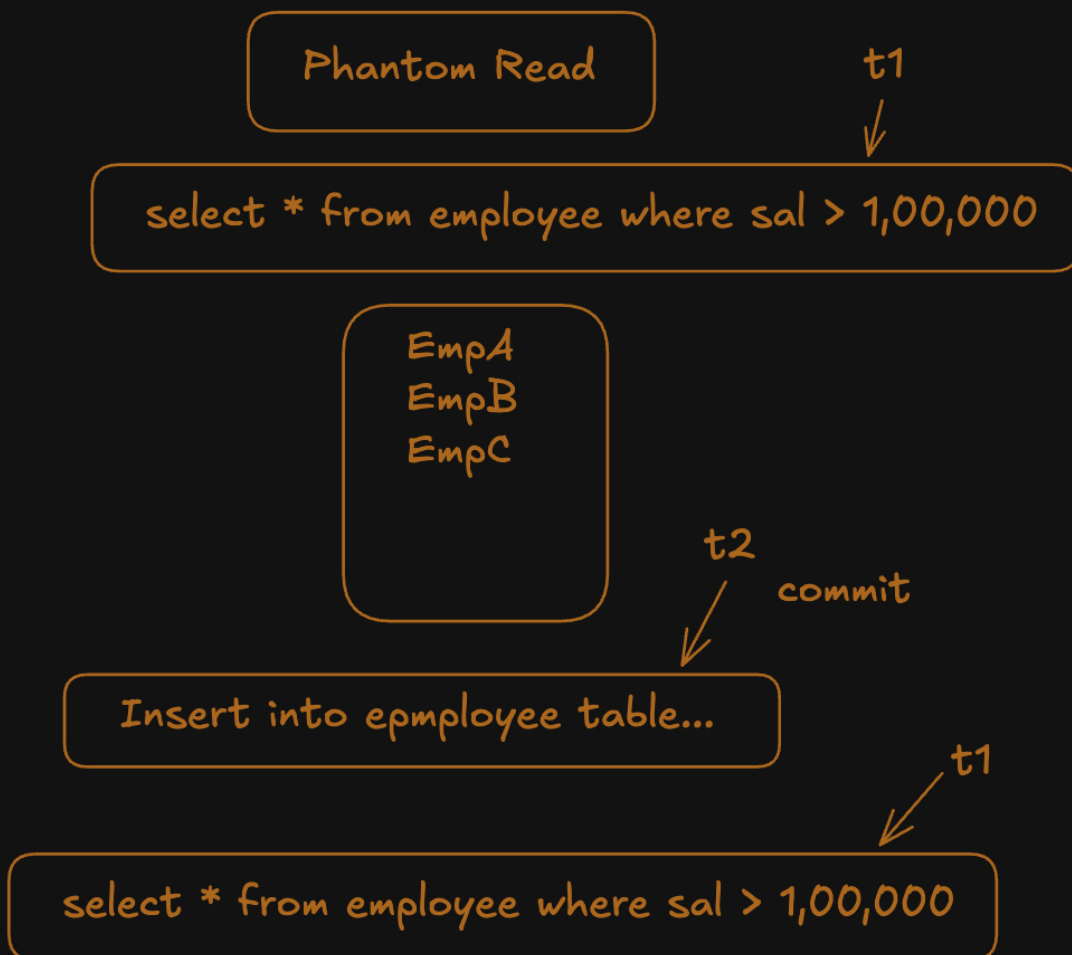
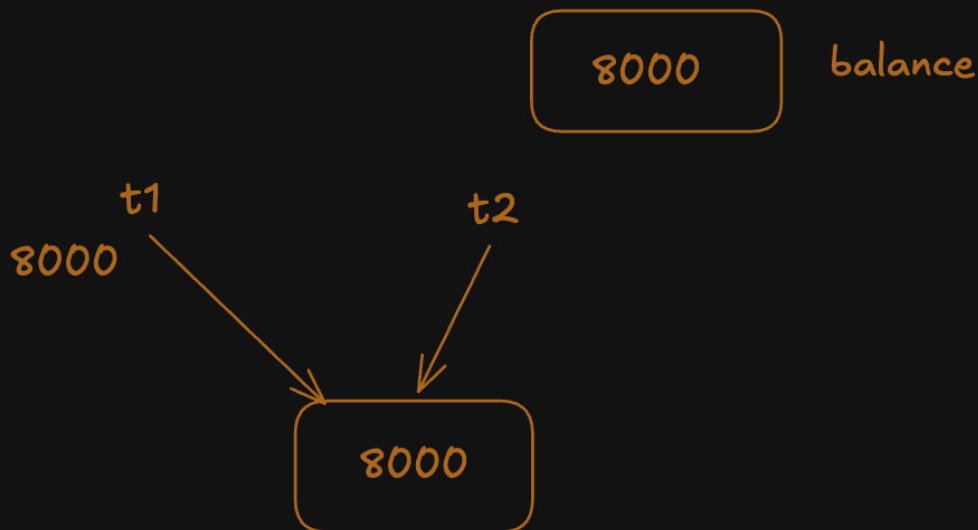
Non-Repeatable Read

txn → 101 Rohit

1 101 Aditya

2 101 Rohit

Time	Transaction T1	Transaction T2
1	Begins	
2	Reads balance = ₹10,000	
3		Begins
4		Updates balance to ₹8,000
5		Commits
6	Reads the same row again	
7	Receives balance = ₹8,000	



EmpA
EmpB
EmpC
EmpD

txn 1

txn2

Isolation.READ_UNCOMMITTED
Isolation.READ_COMMITTED
Isolation.REPEATABLE_READ
Isolation.SERIALIZABLE

concurrency
consistency

Phantom read

READ_UNCOMMITTED ->

Dirty Read
Non-repetableread
Phantom read

Non-repetableread
Phantom read

